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Arbitrum Foundation Grant Application – Cheesecoins Protocol (v2)

Project: Cheesecoins Protocol

Flagship implementation: Nubians North NFT collection

Token: CURD (denomination of the Cheesecoins protocol)

Network: Arbitrum One (live since March 25, 2026)

Total ask: \$150,000 USD – Advanced Growth track, milestone-metered

Applicant: Gregory Garner – solo founder, Nubians North farm (Ontario, Canada)

Repository: <https://github.com/garnergregg/cheesecoins-protocol>

Web: cheesecoins.com · nubiansnorth.com

Cheesecoins is the first agricultural supply-chain settlement protocol on Arbitrum – every link in the chain (vendor, producer, processor, distributor, merchant) is on-chain, paid in CURD, and reconcilable in real time. Live on mainnet. Working dairy goat farm at the center.

What’s novel – productive-capacity tokenization

Food sovereignty and financial freedom are the same idea said two different ways. Both are about escaping dependence on systems that can fail you – industrial food supply, or centralized banking. Cheesecoins is the protocol that lets capital flow toward real productive activity (farms, processors, distributors, retailers) the way banks have always made it flow – but cheaper, faster, and without the gate at a banker’s desk.

The agri-food sector remains structurally under-served in DeFi/RWA tokenization. To the best of our survey of the 2026 landscape, existing platforms either tokenize past harvests as commodity-stablecoins (e.g. Agrotoken on Algorand, ~\$70M transaction volume – a credible commodity-receipt model, but limited to 1-token = 1-ton-of-grain redemption with no NFT layer or supply-chain integration), structure as RWA credit / lending platforms (Goldfinch, Centrifuge – debt-style investor relationships against real-world collateral, no NFT-gated staking), or focus on production-backed

yield outside agriculture (Ayni Gold tokenizes Peruvian gold mining production – closest in design philosophy, but in a different sector and without NFT-as-only-gate). Cheesecoins is structurally different from each.

Four design elements that, combined, do not appear elsewhere in the landscape:

1. **NFT ownership as the only gate** to staking, governance, and yield. No staking, no governance weight, no fractional pool participation, no Super Holder rewards exist without an NFT. Every CURD-holder seeking yield is also a project supporter – by construction.
2. **Productive-capacity tokenization, not asset or commodity-receipt tokenization.** Each NFT represents a claim against the future cash flow an operating business will generate as it scales – closer to a productive-capacity bond than a fractional-equity ownership token. Holders extend capital against growth, not against a static asset or a warehouse-stored commodity.
3. **Full agricultural supply-chain integration via on-chain registry.** MerchantRegistry V3's five-tier enum (vendor / producer / processor / distributor / merchant) means the same NFT-against-productive-plan structure works for a feed supplier, a cheese plant, a regional distributor, or a retail co-op. Capital flows to growth at every link; CURD settles every transaction; yield routes back to NFT holders.
4. **Bank-replacement service catalogue roadmap.** The same primitive scales beyond commerce settlement to equipment credit, insurance, HACCP verification, planning tools, and accounting – all framed as protocol services on the same NFT-against-productive-plan structure. (Roadmap context in docs/PROTOCOL_THESIS.md.)

Each individual element above has prior art in DeFi or RWA. The integrated design – to the best of our survey – does not have a direct comparator in the agri-food sector or beyond.

Each Scene NFT in the Nubians North collection is, simultaneously:

- **A claim on a real-world asset** – a slice of a registered Producer in MerchantRegistry V3: a working dairy goat farm in Ontario generating real cash flow.
- **A yield-bearing financial instrument** – stake CURD against the NFT to earn protocol yield from BootstrapYieldPool and (post-Phase 2) FX-hedge premium reserves and T-bill yield on treasury USDC.
- **A governance key** – staking enough CURD confers Super Holder status, granting log-weighted votes on protocol upgrades and supply expansion (66% supermajority, 30-day vote, 2-day timelock).
- **A marketplace access token** – Phase 2 fractional staking pools wrap each NFT into a public deposit pool that lets anyone DCA into a real working farm in \$5 increments (see §4.4).

The deployment side scales via a **project factory**: the deployed CommerceNFTTemplate proxy lets any registered producer launch their own project NFT collection through a single factory call. Today that's Nubians North. The next hundred farms onboarded under this grant inherit the same battle-tested template – no new contract development required to scale supply.

What's standing today vs what's still ahead:

Today (live on Arbitrum One)	This grant funds	Post-grant roadmap
MerchantRegistry V3 (5-tier supply chain), NubiansNorthNFT (100 of 50,000 minted), StakingManager (NFT-gated, 100 CURD min), BootstrapYieldPool funded with 5M CURD, CurdDirectSale loaded with 10M CURD, full deployment under 2-of-2 Gnosis Safe, 859 Foundry tests passing	30 partners across 5 tiers, CURD/USDC liquidity pair, FX-pegged redemption mode, Settlement Index seed (CME data integration), Real Return Dashboard, fractional staking pools	Real-world-asset NFTs (small lift atop the existing primitive – same gate, new asset class), Farm Planner (web app for businesses to submit plans -> NFT issued against the plan), agricultural finance services (equipment loans, operating credit, insurance, HACCP, accounting – bank-replacement layer per the original tokenomics opportunity model), Food Box consumer integration (more complex; deferred)

This is the protocol’s structural answer to the question “why does anyone hold this NFT instead of any other?” – every utility compounds the others, every new partner registered makes the whole stack more economically meaningful, and the same primitive scales from one farm to a fully-integrated agricultural finance ecosystem.

Full positioning detail in [docs/PROTOCOL_THESIS.md](https://docs.cheese.co/PROTOCOL_THESIS.md).

1. One-paragraph summary

Cheesecoins is an agricultural finance protocol that **tokenizes productive capacity** – issuing NFTs against verified business plans, settling supply-chain commerce on-chain in a single token (CURD), and routing yield from productive activity back to the NFT holders who extend the capital that funds growth. The function being performed is agricultural banking – capital provision against verified productive intent in exchange for yield – moved from gated bank desks to a public on-chain protocol. Today the focus is productive-capacity NFTs (Nubians North as flagship); the same primitive extends naturally to real-world-asset NFTs and a full bank-replacement service catalogue (equipment loans, insurance, HACCP, planning, accounting) on the post-grant roadmap. The protocol is live on Arbitrum One: twelve contracts deployed under a 2-of-2 Gnosis Safe; 200 million CURD minted across treasury, yield pool, and direct-sale contracts; the BootstrapYieldPool holds 5,000,000 CURD for staker yield; CurdDirectSale is loaded with 10,000,000 CURD. As of May 4, 2026, **MerchantRegistry V3 is live on mainnet with a five-tier supply-chain enum** – making Cheesecoins the first protocol to formalize the full agricultural supply chain on-chain, not a generic merchant rail. This grant funds five workstreams measured against verifiable on-chain milestones: a six-month promotional onboarding campaign aimed at registering 30+ supply-chain partners across multiple tiers; a CURD/USDC liquidity pair; an FX-pegged redemption mode for cross-border merchants; the seed integration capital for an on-chain agricultural settlement index that fills a structural gap in DeFi data infrastructure; and consumer-facing additions (NFT-backed fractional staking pools and a real-return dashboard) that lower the user entry point and surface the protocol’s economic story to a broader audience.

2. Mission: food sovereignty as financial freedom

Food sovereignty and financial freedom are the same idea said two different ways. Both are about escaping dependence on systems that can fail you – industrial food supply, or centralized banking. The people who feel that deeply already live the values Cheesecoins is built on. Real things – soil, animals, labour, community – have value that no bank can print away.

Agriculture is consolidating. Small and mid-sized farms are losing access to capital, distribution, and price discovery – not because they produce worse food, but because the financial and supply-chain rails were never built for them. Cheesecoins is a freedom-tech response: rails that level the field without forcing farms to choose between joining and surviving.

Position: **more farms, not fewer.** The protocol is intentionally inclusive of larger operators – they participate as vendors, distributors, or merchants under the same registry as any small farm. The aim is decentralization without alienation.

3. What is live on Arbitrum One today

All of the following are deployed on Arbitrum One mainnet, owned by the Gnosis Safe `0x6C64ACd0Be573D7c90d9b0c6fFDf2E69573871D2`, and verifiable on Arbiscan. Full deployment manifest at `contracts-v2/deployments/42161-arbitrum-one-all.json`.

Each address below links to Arbiscan for one-click verification.

Contract	Address	Status
CheesecoinsCore V2 (CURD token, 200M supply)	0x833551...fB89cD	Live
NubiansNorthNFT (100 unique scenes, 500 copies each, 50,000 max supply)	0x4a99b2...808aAD	Live; 100 minted to date
StakingManager	0xcfb9d9e...cb35B6	Live
BootstrapYieldPool	0x27b55A...9DE68C	Funded with 5M CURD
CurdDirectSale	0x31ea59...12d10D	Loaded with 10M CURD
ProjectSale	0x07FC04...71E7fc	Live (no projects registered yet)
CommerceNFTTemplate (impl)	0xAe2fe1...66dabf	Live
MerchantRegistry V3 (5-tier supply-chain enum)	0xCA7f73...8709bA	Live; Nubians North registered May 4 2026
MerchantSettlement	0x94AA0B...da5FE0	Deployed (config pending)
FiatRedemptionVault	0x9c293c...737274	Deployed (funding pending)
TreasuryRateAdvisor	0x53aB92...1Ec77f	Live
ProtocolTimelock	0x0E7d11...9ebb5C	Live

Treasury verifiable on-chain right now: the Safe holds 165,000,000 CURD; the yield pool holds 5,000,000 CURD; the direct sale holds 10,000,000 CURD. Total accounted: 180M of the 200M supply.

Audit posture: in-house Foundry test suite under contracts-v2/foundry-test/ – **859 tests passing** across all production contracts (CURD core, NFT, staking, sale, commerce, monetary, governance, options). Static analysis via **Slither** (most recent run May 2 2026) – output triaged against the manual audit findings; pattern-flag false positives (mock-contract immutables, non-Reentrant-protected external calls following the checks-effects-interactions pattern, bounded administrative loops) documented and accepted. Manual contract-by-contract security audit (contracts-v2/foundry-docs/SECURITY_AUDIT_REPORT_2026-03.md) – 27 substantive findings identified; all critical-through-low severity remediated and verified, two info-severity items deferred with rationale. Mainnet-fork simulation required pre-broadcast for any state-changing upgrade. Third-party audit will be sought via the **Arbitrum Audit Program** (separate \$10M ARB pool) – see Section 7.

3.1 The 5-tier registry – what it actually does

Every economic actor in the food chain registers in one on-chain registry, with a tier enum:

Tier	Role	Real example
Vendor	Sells inputs / raw materials to a producer	Feed supplier, equipment vendor
Producer	Turns raw inputs into a primary product	Greg / Nubians North
Processor	Adds value (cheese, packaging, abattoir)	Local cheese plant
Distributor	Bulk movement to retailers	Wholesaler
Merchant	Sells to end consumers	Retailer, restaurant, market stall

The registry was upgraded from V2 to V3 on May 4, 2026 – single-tx Gnosis Safe upgrade, mainnet-fork-simulated pre-broadcast, both pre-existing merchants preserved with default `Tier.Merchant`. The full chain Greg already operates day-to-day (Graham Farms -> Nubians North -> cheese plant -> wholesaler -> retail) is now expressible on-chain in a single registry, with tier-aware payment routing through `MerchantSettlement`.

This is the headline differentiator. Anyone can ship a “merchants accept token” rail. Cheesecoins models the **whole chain**.

4. What is proposed and grant-funded

We are explicit about the line between what exists and what the grant funds. Five workstreams sit downstream of the live-on-mainnet base:

4.1 Partner onboarding across all five tiers – promotion-led, the headline grant deliverable

Live registry plus a working POS phone app on cheesecoins.com is the technical foundation. The grant funds **the activity that turns a deployed registry into a live commerce network**: in-person partner onboarding, gas subsidies for first-time merchants, materials in English and French, and a follow-on rollout into the US South via existing operator contacts in Georgia and Mississippi.

4.2 FX-pegged redemption mode – Phase 2 deliverable

Cross-border ag commerce (Canada -> US) inherently has FX risk. A vendor accepts CURD when CAD/USD is at one rate and cashes out at another, the protocol gets blamed for FX losses that aren't its fault. The proposed FxPeggedRedemption mode pegs the merchant's redemption value to the USD/CAD rate at receipt time (read via Pyth, already integrated in the Sepolia options market), funded by a small per-tx premium to a stability reserve. Merchants get **CAD-stable settlement automatically**; the protocol earns predictable revenue from the premium spread; the reserve overflow flows back to NFT holders. Built and audited under the grant; mainnet deploy after audit clears.

4.3 Cheesecoins Agricultural Settlement Index – Phase 2/3 deliverable

The CheesecoinsOptionsMarket is live on Sepolia (corn, wheat, soybeans, USD/CAD) but **NOT being shipped to mainnet on the current data quality**. Reason: USDA AMS publishes weekdays only – Friday-close prices are stale through Monday, which is fine for testnet validation but unacceptable for real-money exercise.

We verified through May 7, 2026 that **no oracle on any chain currently provides reliable agricultural settlement prices**. Pyth Network's 3,000+ price feeds cover crypto, equities, FX, precious metals, oil, and macroeconomic indicators but include zero agricultural commodities – Pyth's own blog frames ag commodities as "potential future additions through governance." Chainlink Data Feeds cover precious metals and oil but not agricultural commodities. RedStone, API3, and Supra advertise broad commodity capabilities but do not publish live agricultural commodity feeds. This is a structural DeFi data gap, not a Cheesecoins-specific limitation.

The grant funds the seed integration of a derived CME-based settlement index – CommodityPrice-Oracle v2 with multi-keeper redundancy, real-time CME-derived ingestion, and a public consumption interface (getPrice(marketId) already deployed). Other DeFi protocols then consume the index; Cheesecoins becomes the on-chain settlement index for agricultural commodities, not a downstream consumer of someone else's feed.

CME outreach is in flight (initial contact sent May 5, 2026); the grant's modest CME line item is seed integration capital while the data-licensing terms are negotiated directly.

4.4 NFT-Backed Fractional Staking Pools – Phase 2 deliverable

The protocol's current staking minimum is **100 CURD per NFT position** (Config.MIN_STAKE_PER_NFT enforced in StakingManager.sol). For young or budget-constrained users – the audience most likely to convert on the "support a real farm + earn yield + revolt against the system that bleeds you" pitch – a \$100 minimum is a meaningful barrier.

The proposed FractionalStakingPool.sol lets every existing Nubians North NFT operate as a yield-bearing deposit pool accepting **micro-deposits as small as \$5 in CURD**. The pool aggregates contributions from many small users, stakes the aggregate against the underlying NFT once the protocol minimum is hit, and distributes earned yield proportionally to depositors. The NFT holder running the pool earns a **small management fee** – turning collection ownership into a recurring revenue stream and creating a real economic incentive to acquire and hold Scene NFTs beyond their initial collectible appeal.

Mechanism: delegated staking via StakingManager V2 upgrade.

The StakingManager proxy (0xcfb9d9e4B97DD40863b134e7979d3038Ec5cb35B6) is upgraded to V2 implementation with a new stakeFor(nftId, amount, lockMonths, onBehalfOf) function callable by NFT-owner-approved addresses. The NFT stays in the original holder's wallet; the FractionalStakingPool contract is approved (per-NFT, revocable) to stake the aggregated deposits on the holder's behalf. Yield distribution honors the original holder's Super Holder status – meaning **the holder keeps governance rights** while their NFT is generating pool revenue.

A custodial-pool alternative (NFT transferred into pool, pool becomes owner) was evaluated and rejected. While simpler to ship, it strips the original holder of Super Holder governance status during the period their NFT backs the pool – a non-trivial concession that misaligns long-term holder economics with governance participation. The delegated design preserves both.

Implementation involves a small upgrade to the existing StakingManager proxy (introducing a delegated-staking entry point so a pool contract can stake on a holder's behalf without taking custody of the NFT), a new aggregating pool contract that batches dollar-scale deposits up to the protocol minimum and distributes yield pro-rata to depositors, full Foundry test coverage of the V1->V2 migration plus pool aggregation and yield distribution, third-party audit via the Arbitrum Audit Program (separate \$10M ARB pool), and a Super NFT governance proposal authorizing both changes (66% supermajority, 30-day vote, 2-day timelock) before execution. The upgrade is purely additive – backwards-compatible at the proxy level – and surfaces a deposit UI on cheesecoins.com once authorized.

Neither side modifies the existing NubiansNorthNFT contract. The work is purely additive: a backwards-compatible StakingManager upgrade + new pool contract.

Practical impact: lowers the user entry point from \$100 to \$5, opens the protocol to the recurring-micro-investment pattern (“DCA \$5/month into a real farm”), and aligns long-term NFT-holder economics with broader user adoption.

4.5 Real Return Dashboard – Phase 2 frontend deliverable

Proposed under this grant – does not yet exist on cheesecoins.com. A single public page at cheesecoins.com/inflation that surfaces the protocol's economic story in plain English: **CURD is dollar-tracking** (USDC-backed); the inflation hedge is realized via staking. That nuance is currently buried in the technical docs.

The dashboard surfaces a real-return number – staker yield minus US CPI year-over-year – pulled from authoritative free public data (Federal Reserve FRED) plus on-chain reads from the protocol. Calibrated to honest math:

“Holding dollars lost you ~3% to inflation last year. Staking CURD earned you ~5% in yield. Real return: +2%. Updated daily.”

Staker yield is one of several economic surfaces in the protocol – alongside commerce fees, FX hedge premium, and (post-CME integration) Settlement Index subscriptions. This dashboard surfaces the consumer-facing one. Full revenue breakdown is in §8; modeled financial projections (P&L, cash-flow, balance sheet, scenarios across Y1–Y3) live in docs/CHEESECOINS_FINANCIAL_PROJECTIONS.xlsx.

The deliverable gives Tier 1A and 1B audiences a concrete sharable artifact addressing the “savings are eroded by holding dollars” pain point, and provides a content engine for the Annie + Alice educational channel – daily inflation-vs-yield commentary writes itself. Built under the grant by extending the existing Next.js front-end; no smart-contract changes required.

5. Why Arbitrum

The protocol's core unit economics – settling small merchant transactions in CURD, distributing micro-rewards across thousands of NFT holders, and producing on-chain food-safety records, supply-chain provenance, and HACCP-grade audit trails across the production chain – only work on a chain where a transaction costs cents, not dollars. Arbitrum's combination of low fees, Ethereum-equivalent security, and the deepest L2 stablecoin liquidity is the only environment where this protocol is economically viable.

Specific Arbitrum-native dependencies: - Native USDC on Arbitrum (0xaf88d065...) as the protocol's on-ramp asset - **Uniswap V3 on Arbitrum** as the venue for the CURD/USDC pair, deployed at the 0.05% stablecoin fee tier (the standard for pegged-pair pools). Chosen for aggregator routing priority (1inch / Matcha / Paraswap / 0x all source Uniswap V3 first), broadest wallet UX, and lowest discovery friction for new buyers - Gnosis Safe on Arbitrum One as the existing treasury custodian - Arbiscan as the merchant- and consumer-facing verification surface - Pyth Network on Arbitrum as the FX oracle for the FX-pegged redemption mode

6. Roadmap and milestones

Grant disbursement is tied to verifiable on-chain milestones. Every trigger reads from public Arbitrum One state – reviewers can independently verify via cast logs or Arbiscan. **Greg has begun partner onboarding in May 2026 independent of this grant** (the protocol is live; partners can register today). The grant accelerates and extends what's already in flight.

All disbursements occur post-M0 (grant signing). Where a milestone trigger is already verifiable on-chain at the time of M0 – for example M1, where 5 partners may already be onboarded by Jul 2026 – that tranche disburses at M0 signing on trigger verification. Pre-grant work counts; pre-grant cash does not flow.

#	Target	Milestone	Trigger (on-chain verifiable)	Disbursement
M0	Jul 2026	Grant accepted, promo campaign launches	Grant agreement signed; promo spend begins	\$55,000
M1	Jul 2026 (state achieved Jun 2026)	5 partners across 2+ tiers	5 setMerchantWithTier events on MerchantRegistry (verified at M0 signing if already on-chain)	\$25,000

#	Target	Milestone	Trigger (on-chain verifiable)	Disbursement
M2	Jul 2026	CME data agreement signed; integration test live on Sepolia	Signed agreement evidence + on-chain price update events on CommodityPriceOracle	\$13,000
M3	Aug 2026	15 partners across 3+ tiers, CURD/USDC pool live	15 MerchantStatusChanged events + Uniswap V3 0.05% stablecoin pool deploy tx	\$35,000
M4	Aug 2026	First Commerce NFT project fully sold out (<i>tracked, informational</i>)	All scenes in a partner Commerce NFT project hit SCENE_MAX_SUPPLY cap	<i>No disbursement; positive demand signal</i>
M5	Aug 2026	FxPeggedRedemption audited and deployed to mainnet	Audit report published + module configured on FiatRedemptionVault proxy	\$17,000
M6	Dec 2026	30 partners across all 5 tiers	30 MerchantStatusChanged events covering all 5 Tier enum values	\$5,000
M7	Mar 2027	Nubians North NFT 50% sellout (<i>tracked, informational</i>)	mintedPerScene cumulative $\geq$ 50% of T0-TAL_MAX_SUPPLY (25,000 of 50,000)	<i>No disbursement; capacity signal – see §8</i>
			Total disbursed	\$150,000

The full per-milestone disbursement breakdown by bucket (promo, liquidity, founder development, operations, contractor, audit, CME) lives in [docs/GRANT_DISBURSEMENT_SCHEDULE.xlsx](#) – editable workbook with milestone dates, disbursement schedule, and repayment-trigger logic.

M6 partner distribution and soft-cost economics. The 30-partner target across all five tiers breaks down as 12 merchants / 7 vendors / 5 producers / 3 distributors / 3 processors – reflecting the realistic Ontario small-ag ecosystem composition. Per-tier rationale, soft-cost components, and the breakdown of how the \$80,000 promotion bucket is spent (paid acquisition vs direct onboarding cost) are documented in [docs/PROTOCOL_THESIS.md](#).

Expected commerce flow. With 30 onboarded partners settling commerce in CURD across the 5-tier supply chain, the protocol expects to drive **\$200,000 to \$1,500,000+ in CURD-denominated commerce volume** (sum of MerchantSettlement event amounts on MerchantRegistry V3) within the first 12 months – an order of magnitude larger than the grant ask, all on-chain and verifiable. Modeled scenarios across Y1–Y3 in [docs/CHEESECOINS_FINANCIAL_PROJECTIONS.xlsx](#).

All triggers are verifiable on-chain or in published artifacts – no subjective milestones.

7. Budget breakdown – \$150,000 total

Promotion-led structure, milestone-gated, on-chain auditable.

Bucket	Amount	%	Release gate
Promotion + partner incentives	\$80,000	53%	Front-loaded – \$50k at M0 kickoff, \$20k at M1 (5 partners), \$10k at M3 (15 partners + pool live)
Founder development	\$18,000	12%	\$3k disbursed at each of M0, M1, M2, M3, M5, M6 – six tranches, milestone-tied
External contractor capacity	\$6,000	4%	M5: hired as needed against FxPegged audit prep + multi-keeper infra deliverables
Liquidity tranche (CURD/USDC seed on Uniswap V3)	\$20,000	13%	M3: released when 0.05% stablecoin-tier pool deploys at calibrated start price
Ongoing operations stipend (multi-keeper infra, uptime, support)	\$12,000	8%	\$2k disbursed at each of M0, M1, M2, M3, M5, M6 – six tranches alongside founder dev
CME data seed	\$8,000	5%	M2: released on signed CME agreement + first integration test live on Sepolia

Bucket	Amount	%	Release gate
Audit + contingency	\$6,000	4%	M5: layers with the separate Arbitrum Audit Program application

Total: \$150,000.

Why this shape: - Founder is solo. Time is the binding constraint, not capital. Most of the ask funds the activity that compounds (promotion + onboarding) instead of one-off infrastructure. - Promotion-as-gate produces verifiable on-chain progress; reviewers can cast logs on the registry instead of reading status reports. - CME at \$8k seed (not a 12-month license commitment) – Greg is negotiating usage-based terms with CME directly. The grant funds initial integration, not perpetual licensing. - Audit is deferred to the Arbitrum Audit Program (separate \$10M ARB pool); the \$6k contingency layer is for findings-remediation gas costs and final attestation.

8. Sustainability and revenue share back to Arbitrum DAO

Cheesecoins is not asking for perpetual subsidy. The grant is the **runway** for a sustainable subscription business; post-grant, the protocol covers its own infrastructure costs from:

1. Protocol fees on commerce (5% commerce / 2.5% capital, baked into ProjectSale)
2. Subscription fees from consumer protocols using the Cheesecoins Agricultural Settlement Index
3. FX hedge premium spread (post FxPeggedRedemption mainnet deploy)

Revenue share back to Arbitrum DAO – capacity-gated.

Repayment commences only when the protocol demonstrates **sufficient financial capacity to repay without compromising operating runway**. Specifically: the repayment clock starts on the **earlier** of:

1. **Nubians North NFT cumulative primary-sale proceeds $\geq$ \$225,000** (= $1.5 \times$ the grant cap). Verifiable from `NubiansNorthNFT.totalSupply()` * primary mint price on Arbiscan. Translates to roughly 6% of the 50,000-supply collection sold out – meaningful demand, but well short of the M7 50% half-mark milestone.
2. **Cumulative CURD-denominated commerce volume $\geq$ \$1,000,000 USD-equivalent**. Verifiable as the sum of `MerchantSettlement` event amounts on `MerchantRegistry V3`. Demonstrates the protocol's commerce flow is at a level where 5% routing to repayment is sustainable.
3. **Settlement Index subscription revenue is live and accruing** – protocol earns recurring revenue from consumer protocols using the on-chain ag price index (post-M2).
4. **Treasury USDC balance $\geq$ remaining repayment obligation** – protocol can repay outright without ongoing fee routing.

Once any threshold trips, the protocol's 5% commerce fee on subsequent NFT and merchant settlement activity (plus any Settlement Index subscription rev-share) routes 5% to the Arbitrum DAO repayment escrow until the cap is hit.

Important: a single partner Commerce NFT project selling out does not by itself trigger repayment. Partner projects can be small (\$40k CSAs, micro-launches) and a sellout there is positive demand but not financial capacity. The thresholds above ensure repayment begins only when the protocol can support it – protecting both the DAO (real money flowing back) and the protocol (operating runway intact).

Cap: cumulative repayment $\leq 1.5 \times$ grant amount (\$225,000), or 36 months from the first threshold trip, whichever comes first. After the cap is met, share drops to 0%.

Why this shape: - Bounded enough to keep the protocol attractive to future investors and lenders - Generous enough to signal serious commitment to fairness (“we’ll pay you back 50% premium when it succeeds”) - Aligned with the zkFetch precedent (3% / 10% revenue share to DAO) but capacity-gated to avoid forcing repayment before the protocol can afford it - Multiple paths mean the DAO gets paid back from whichever surface of the protocol moonshots first – Nubians North primary demand, broader CURD commerce flow, Settlement Index subscriptions, or treasury accumulation

9. KPIs reported quarterly

All metrics derivable from on-chain events. No self-reported numbers.

- Partners registered, broken down by Tier enum value (MerchantRegistry MerchantStatusChanged + MerchantTierSet events)
- Total settlement volume (sum of MerchantSettlement event amounts)
- CURD/USDC pool depth and 30-day volume
- Active wallets paying via the merchant rail (90-day rolling)
- Commerce NFT projects launched and sold (ProjectSale events)
- Settlement Index oracle uptime % and freshness lag (post-M2)
- NFT secondary market activity on Nubians North scene collection

10. Geographic rollout

Phase 1 – Ontario, Canada: founder’s home jurisdiction. Existing partner relationships across all five tiers (raw milk supplier Graham Farms, the cheese plant downstream, Nubians North as producer, regional retailers and direct-to-consumer markets). Lower-friction onboarding for the first 5–15 partners.

Phase 2 – US South (Georgia and/or Mississippi): the founder has existing operator contacts in both states. Georgia has a strong dairy and direct-to-consumer farm-market corridor; Mississippi has ag commerce volume but is underserved by fintech. Cross-border CA->US ag commerce naturally introduces the FX risk that the FxPeggedRedemption mode (Section 4.2) exists to solve – the geographic plan and the technical roadmap reinforce each other.

11. Risk and honesty disclosures

- The protocol is **founder-controlled at Stage 0 governance**. Decentralization is gradual and already wired on-chain: the deployed FounderDecentralization contract reduces founder governance weight on a 5-year linear schedule (50% Y1 -> 40% Y2 -> 30% Y3 -> 20% Y4 ->

10% Y5 -> 0% cliff at Y5+). Super NFT holders gain proportional weight. This is documented in docs/TOKENOMICS_V2.md §3.3 and verifiable on Arbiscan.

- **Founder allocation is locked in 4-year linear vesting.** The full 20M CURD founder allocation was transferred into FounderVestingWallet (0xbac3d40668Ce4030ab5D8cF0bBCFDA457E1216f5) on May 5, 2026 with start = March 25, 2026 (original mainnet launch date) – restoring the originally-designed vesting schedule that was authored at launch but not deployed. As of grant submission, ~2.8% (~575k CURD) is vested; the rest remains locked. Verifiable on-chain: `cast call <vesting> "releasable()(uint256)"`.
- One operations wallet was compromised on March 28, 2026 via an EIP-7702 delegation attack during an unrelated ENS interaction (~\$130 lost). Mainnet was never exposed; the Safe was cleaned the same day. Sepolia testnet contracts deployed by that wallet are quarantined; the Sepolia stack was redeployed under a clean key on May 2, 2026 and the V3 registry upgrade was tested end-to-end on Sepolia before mainnet. Full incident notes in repo memory.
- The protocol carries **no debt and has no outside investors**. The grant is the first external capital.
- **Mainnet options market is intentionally NOT shipping yet.** The current Sepolia oracle (USDA AMS keeper) is fine for testnet but stale on weekends – unacceptable for real-money exercise. Phase 2 deliverable, contingent on the CME data integration. We’d rather miss a ship date than expose users to weekend-stale settlement.

12. Team and growth structure

Founder (current). Gregory Garner – solo founder. Operates Nubians North dairy goat farm in Ontario, Canada. Has been building the Cheesecoins protocol since April 2025 – 13 months of solo work as of submission. Solidity / Foundry on the contract side; Next.js + Thirdweb on the front-end at cheesecoins.com. Treasury secured via 2-of-2 Gnosis Safe (Ledger + MetaMask). Documentary on the project (“The Money That Feeds You”) on YouTube.

Advisory committee (forming). A three-person advisory committee is being assembled to provide strategic counsel through the grant period. Compensation is structured as a CURD treasury allocation with vesting (no grant funds used for advisor compensation). Target areas of expertise: (a) agricultural commerce and supply-chain operations, (b) Solidity / smart-contract security, (c) DeFi go-to-market and partner growth. Names will be published before milestone M3 disbursement (15 partners + CURD/USDC pool live).

Contractor capacity (reserved). \$6k of the grant budget is held against specific M4-stage deliverables (FxPegged audit prep, multi-keeper oracle infra). Engaged on a per-deliverable basis, not as a standing salary. This keeps the operation lean while providing structured surge capacity when audit-grade work is in flight.

Parallel ecosystem support. We will apply to the **Arbitrum Mentorship Program** Cohort 2 when applications open. The protocol matches the program’s eligibility profile (early-stage builder, working prototype on Arbitrum, MVP/seed stage). Mentorship Program engagement is independent of this grant – no overlap or double-funding.

Background – the founder.

Greg grew up on a registered Holstein dairy farm but couldn’t buy his way into Canada’s quota-locked dairy system as a young farmer – so he had to move on from that dream. He earned an Ag Production and Management degree from University of Guelph’s Ridgetown Campus, worked

breeding-stock exports for Rowntree Farms and Cormdale Genetics, ran Manderley's turfgrass operation for five years, and then built Elevated Landscape Technologies – a green-roof and living-wall company the 2008 housing crisis nearly destroyed.

The years he spent fighting banks to recover from 2008 – and refusing to file bankruptcy, despite the pressure to do so – taught him how the monetary system actually works. One banker said it to his face: *“we don't need to work with you – we need a certain number of bankruptcies a year anyway.”* Greg never became one of those statistics, but the line stuck. That sent him down the rabbit hole. He started following Bitcoin, watched the protocol design space mature, and eventually came back to dairy through goats – specifically because goats sit outside the quota system that had locked him out as a kid. Cheesecoins is the synthesis of that journey: food, freedom, and farming combined into a protocol designed as both an exit strategy from a system that pulls the rug out from under small operators on schedule, and a wealth-builder for the next generation that's tired of being told to wait their turn.

Father, Husband and now Grandfather, Greg has a vested interest in effecting change and creating an environment where financial freedom (same as Freedom) can be a reality for generations to come. When Greg is not busy trying to explain blockchain to the huddled masses he is busy being a Husband, Father, Grandfather, and enjoys playing guitar, playing hockey, working on the farm, and of course cooking a nice meal to enjoy with Cheryl over a bottle of wine.

That dinner-table image is also Cheesecoins' path to mass adoption. Most people will never read a DeFi whitepaper – but everybody eats. The protocol meets them where they already are: at the farmers' market, the local cheese counter, the family dinner table. The blockchain becomes invisible plumbing; the food, the farm, and the community do the convincing. That is how Cheesecoins onboards the people still on the outside looking in – not by explaining DeFi to them, but by giving them something real to belong to.

13. Verifiable links

On-chain (Arbiscan): - Treasury Safe (2-of-2 Gnosis): <https://arbiscan.io/address/0x6C64ACd0Be573D7c90d9b0>
- CURD token: <https://arbiscan.io/address/0x833551C5433551fDA5b49D03044D7Df51ffB89cD> -
MerchantRegistry V3 (verified): <https://arbiscan.io/address/0x033435CA67B79aCe3d9052157c34a584a5A8FDE>
- FounderVestingWallet (verified): <https://arbiscan.io/address/0xbac3d40668Ce4030ab5D8cF0bBCFDA457E1216>
- FounderDecentralization (governance schedule): <https://arbiscan.io/address/0xC2378eC98B8Aedf8E748b86775>

Repository: - Source code: <https://github.com/garnergregg/cheesecoins-protocol> - Main-net manifest: <https://github.com/garnergregg/cheesecoins-protocol/blob/main/contracts-v2/deployments/42161-arbitrum-one-all.json> - Tokenomics V2 spec: https://github.com/garnergregg/cheesecoins-protocol/blob/main/docs/TOKENOMICS_V2.md - Financial projections workbook: https://github.com/garnergregg/cheesecoins-protocol/blob/main/docs/CHEESECOINS_FINANCIAL_PROJECTIONS.xlsx

Public: - Live front-end: <https://cheesecoins.com> - Flagship NFT project: <https://nubiansnorth.com>
- Documentary (“The Money That Feeds You”): <https://youtu.be/fqV4zsG7Go>